

How can warehouse strategies boost ROI in modern supply chains?

In today's supply chain landscape, efficient warehouse operations are vital for businesses aiming to boost ROI.

- **Demand forecasting:** Efficient warehousing hinges on precise demand forecasting. By analysing customer demand patterns, companies like Zara adjust stock levels in real time, reducing waste and carrying costs while minimising overstock to maximize ROI.
- **Warehouse layout optimisation:** An efficient warehouse layout reduces travel time, labour costs, and processing times. Toyota uses lean principles to streamline its layouts, cutting waste and improving flow, thereby maximising ROI.
- **Supply chain visibility:** Enhanced supply chain visibility optimises warehousing through real-time tracking of inventory and orders, improving decision-making, and reducing lead times. Coca-Cola's network connects all supply chain parties, enhancing data sharing for better production and inventory management.

By implementing these strategies effectively, companies can reduce operational costs, improve productivity, and enhance customer satisfaction.

tect hazards like spills and blocked aisles, using cameras as early-warning systems. This approach ensures safer working conditions by monitoring vehicle safety and personal protective equipment (PPE) usage, aiming for zero injuries. Meanwhile, Canadian startup AiDANT offers behaviour recognition solutions to improve employee security through products like AiDANT SPARK for vehicle detection and time monitoring, AiDANT AWARE for loitering and intrusion detection, and AiDANT CARE for fall detection and occupancy monitoring. These vision-based systems deliver critical behavioural insights and real-time alerts, enhancing operational security and employee safety in warehouses.

6. Data-driven workforce optimisation. Workforce optimisation involves managing labour training, development, and performance. Using data analytics and software tools helps warehouses improve labour utilisation and reduce costs. Technologies like warehouse and labour management systems enhance training and support by identifying employee needs. Augmented reality (AR) provides real-time inventory and order fulfilment information, enabling faster and more accurate operations. Clear performance

metrics and regular feedback help businesses understand worker expectations and boost productivity. HapiGig, a US-based startup, offers flexible warehouse staffing solutions, reducing hiring costs by providing skilled, on-demand workers. This platform supports warehouse managers by supplying experienced labour to handle volume fluctuations, callouts, and last-minute workforce needs.

7. IoT and digital twins. Startups in the warehousing sector are leveraging IoT technologies to improve process control and optimisation. These systems use connected sensors to track assets and monitor environmental conditions like temperature, humidity, and lighting, facilitating efficient inventory management through real-time data from IoT beacons. Additionally, smart shelves with IoT capabilities provide constant stock updates and alert when replenishment is needed.

A notable advancement is the use of digital twins, which create dynamic, real-time models of warehouse operations from data integrated across warehouse networks and management systems. This technology allows managers to visualise operations, track inventory, and simulate layout and workflow

adjustments effectively.

Indian startup ScratchNest is advancing this field with compact IoT beacons that use Bluetooth low energy (BLE) for precise tracking of inventory and personnel, boasting long battery life. In the US, Celona specialises in 5G local area network solutions, establishing private 5G networks that enhance warehouse automation and support autonomous robots and vehicles, thereby reducing costs and improving operational uptime.

8. Immersive reality. Efficient delivery within warehouses depends on both sophisticated equipment and skilled personnel, prompting an increased use of immersive reality technologies. Virtual reality (VR) and augmented reality (AR) enhance training, planning, and operational efficiency. VR immersively trains employees on various tasks, while AR assists in navigating efficient routes for product retrieval, thus optimising delivery processes and aiding decision-making by visualising operations comprehensively.

UK-based startup EHS VR offers VR forklift training that monitors progress and provides personalised instruction without the need for physical equipment. Canadian startup FlashQ's virtual waiting AI for trucking logistics reduces congestion, saving on fuel costs and minimising environmental impact, benefiting the industry both economically and environmentally.

9. Advanced security. Security is paramount in warehouses due to the high risk of theft and breaches. Warehouses implement robust security measures, including alarms, video surveillance, smart locks, and broken glass detectors. Many are moving to cloud-based systems for secure data transmission and investing in cybersecurity to protect against data fraud. Blockchain technology is increasingly used for enhancing transparency in supply chain transactions.